

Kristin Jacobs Coral Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network** and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

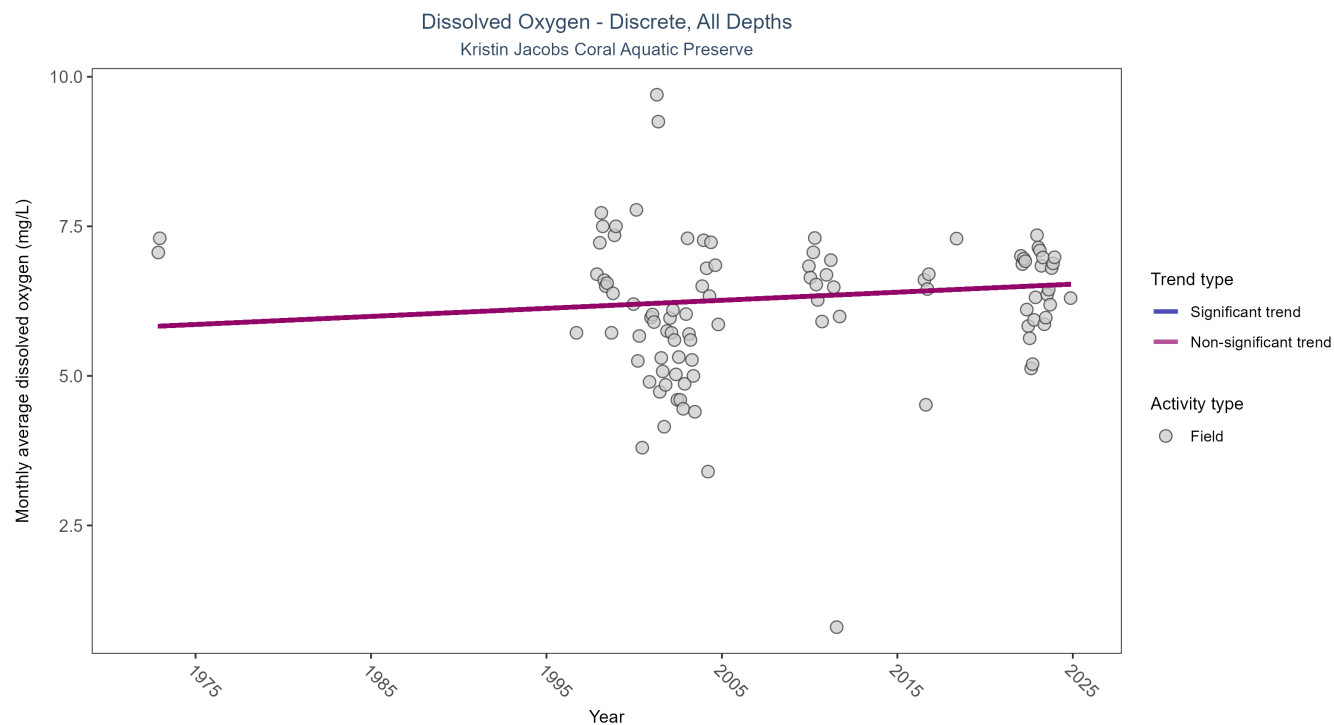


Figure 1: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	715	18	1972 - 2024	6.5	0.1059	5.8188	0.0135	0.3679

Dissolved oxygen showed no detectable trend between 1972 and 2024.

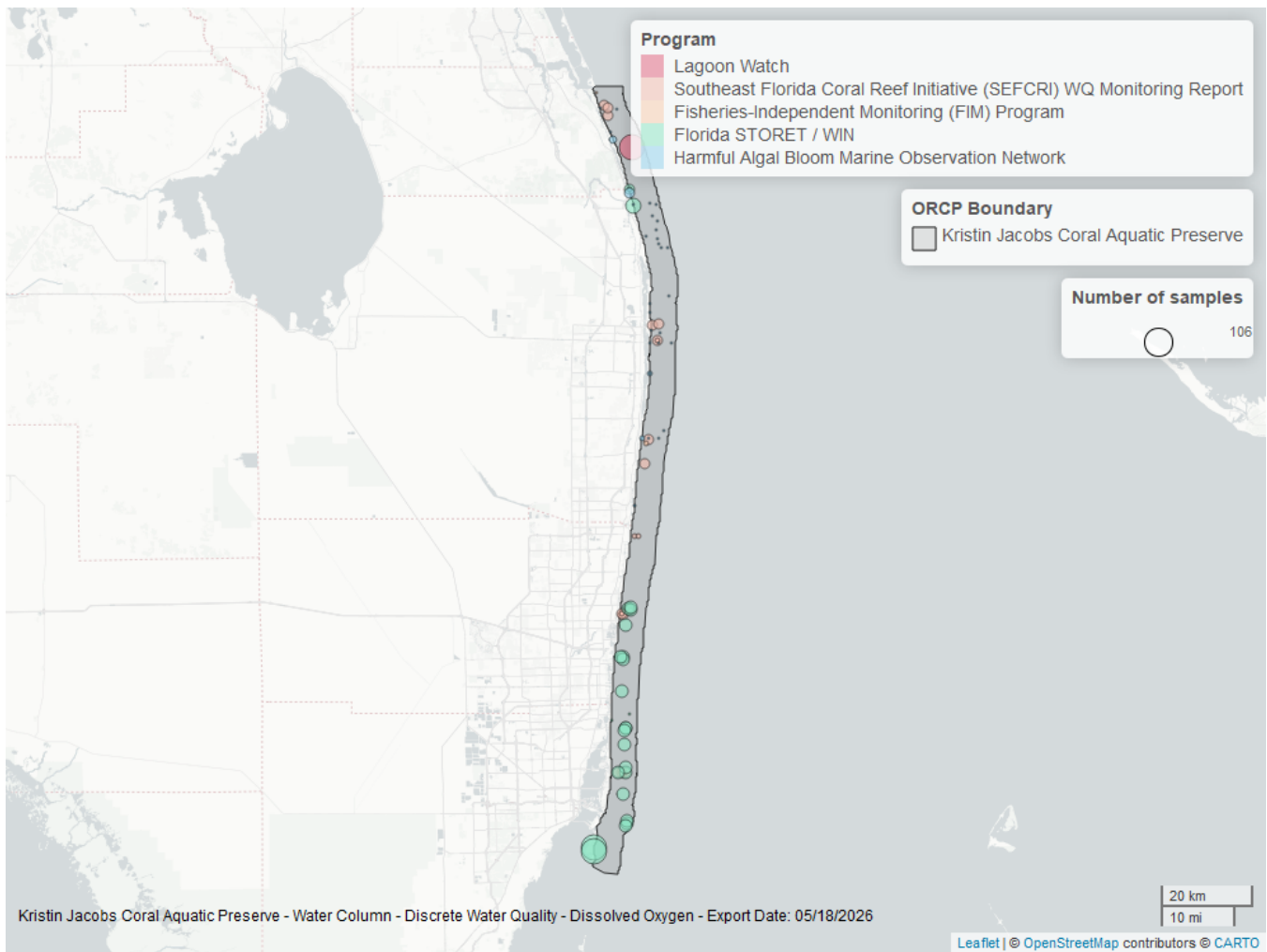


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	786	1996	2023
5058	266	2009	2011
3001	98	1999	2003
95	69	1972	2018
69	5	1998	2024

Program names:

5002 - Florida STORET / WIN¹

5058 - Southeast Florida Coral Reef Initiative (SEFCRI) Water Quality Monitoring Report²

3001 - Lagoon Watch³

95 - Harmful Algal Bloom Marine Observation Network⁴

69 - Fisheries-Independent Monitoring (FIM) Program⁵

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

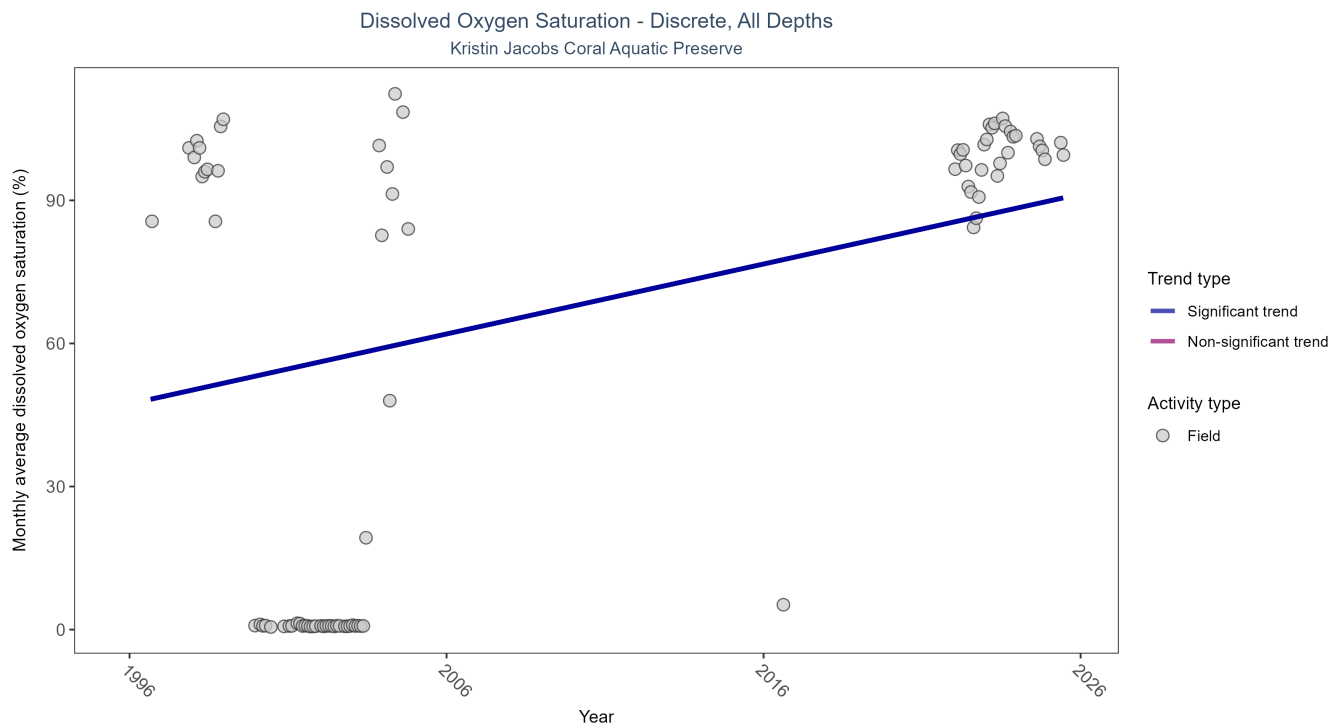


Figure 3: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	1418	14	1996 - 2025	101.5	0.3764	47.3187	1.4668	0.0001

Monthly average dissolved oxygen saturation increased by 1.47% per year.

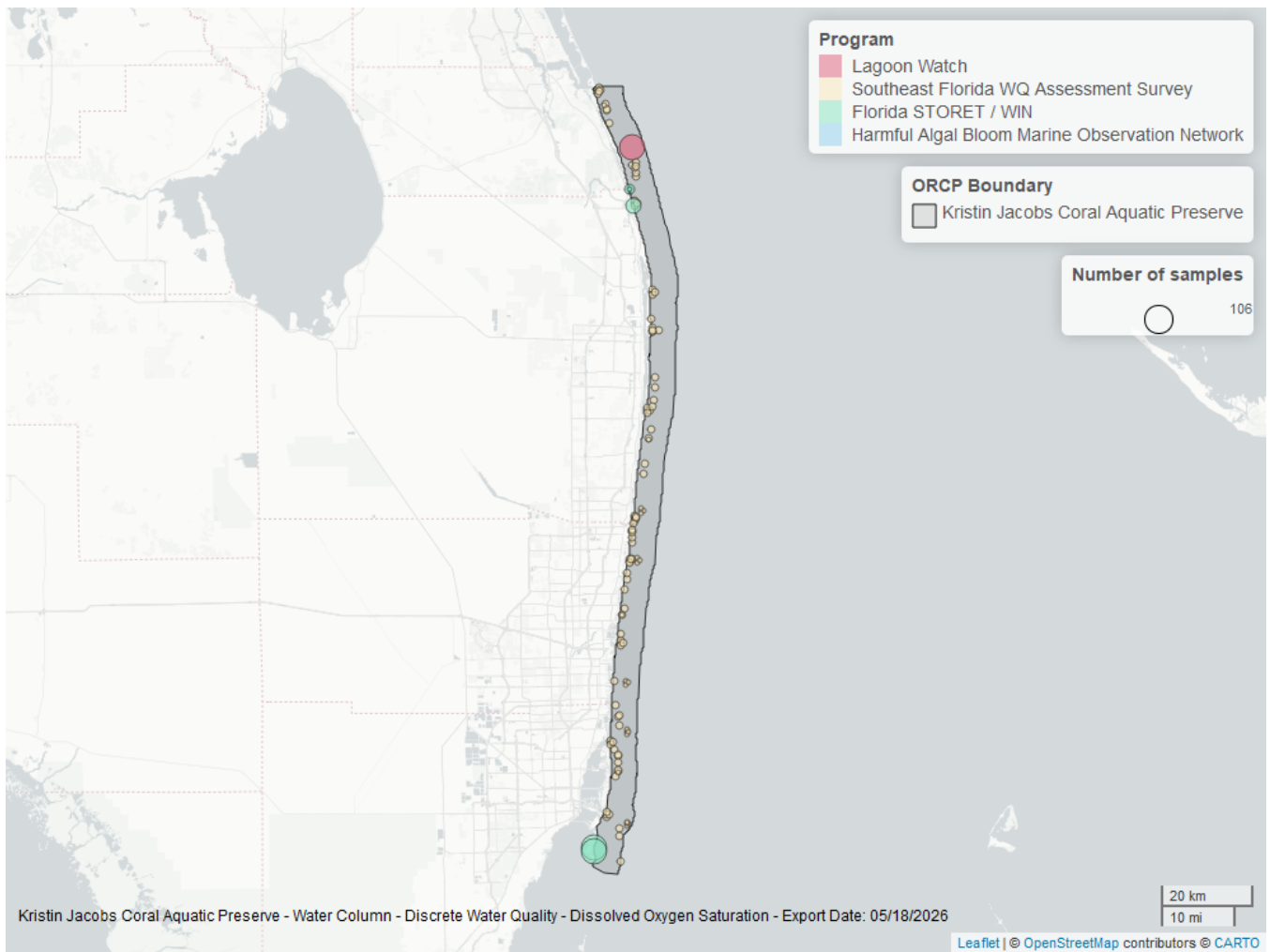


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
5033	1042	2024	2025
5002	277	1996	2023
3001	98	1999	2003
95	1	2016	2016

Program names:

5033 - Southeast Florida Water Quality Assessment Survey⁶

5002 - Florida STORET / WIN¹

3001 - Lagoon Watch³

95 - Harmful Algal Bloom Marine Observation Network⁴

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

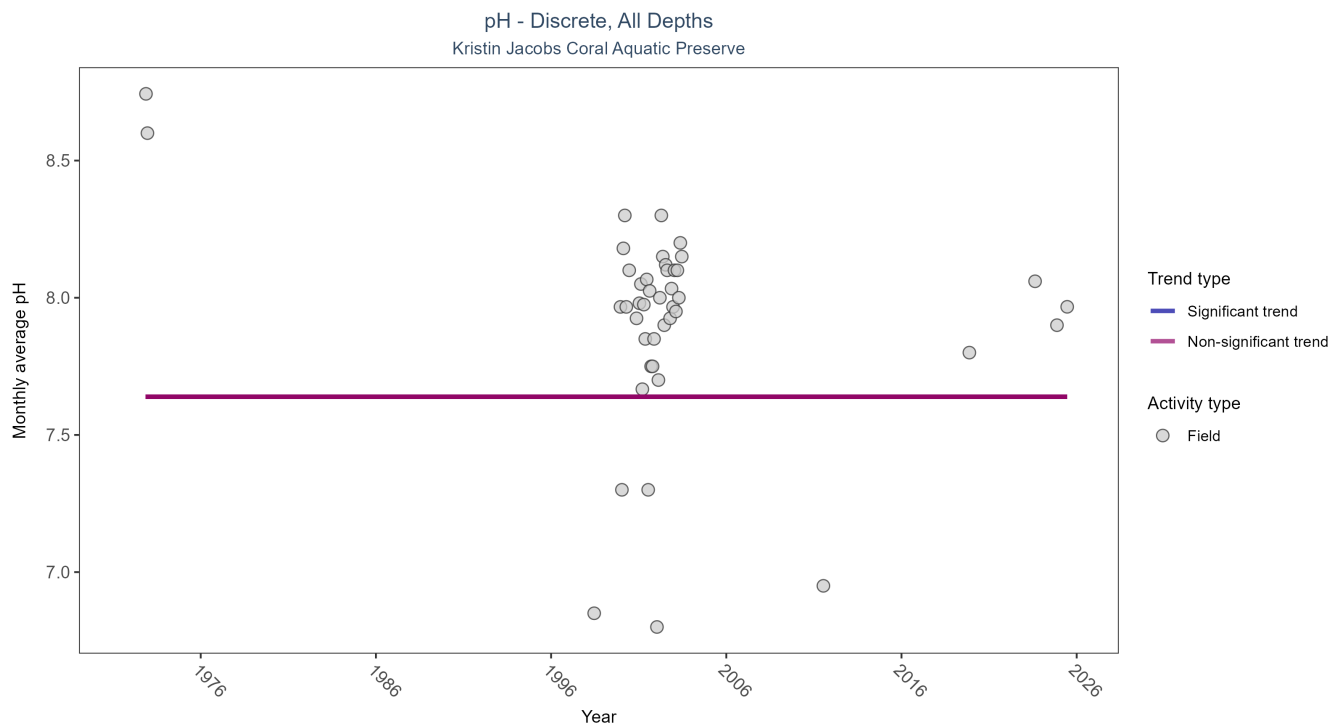


Figure 5: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	376	12	1972 - 2025	7.99	0.0651	7.6392	0	1

pH showed no detectable trend between 1972 and 2025.

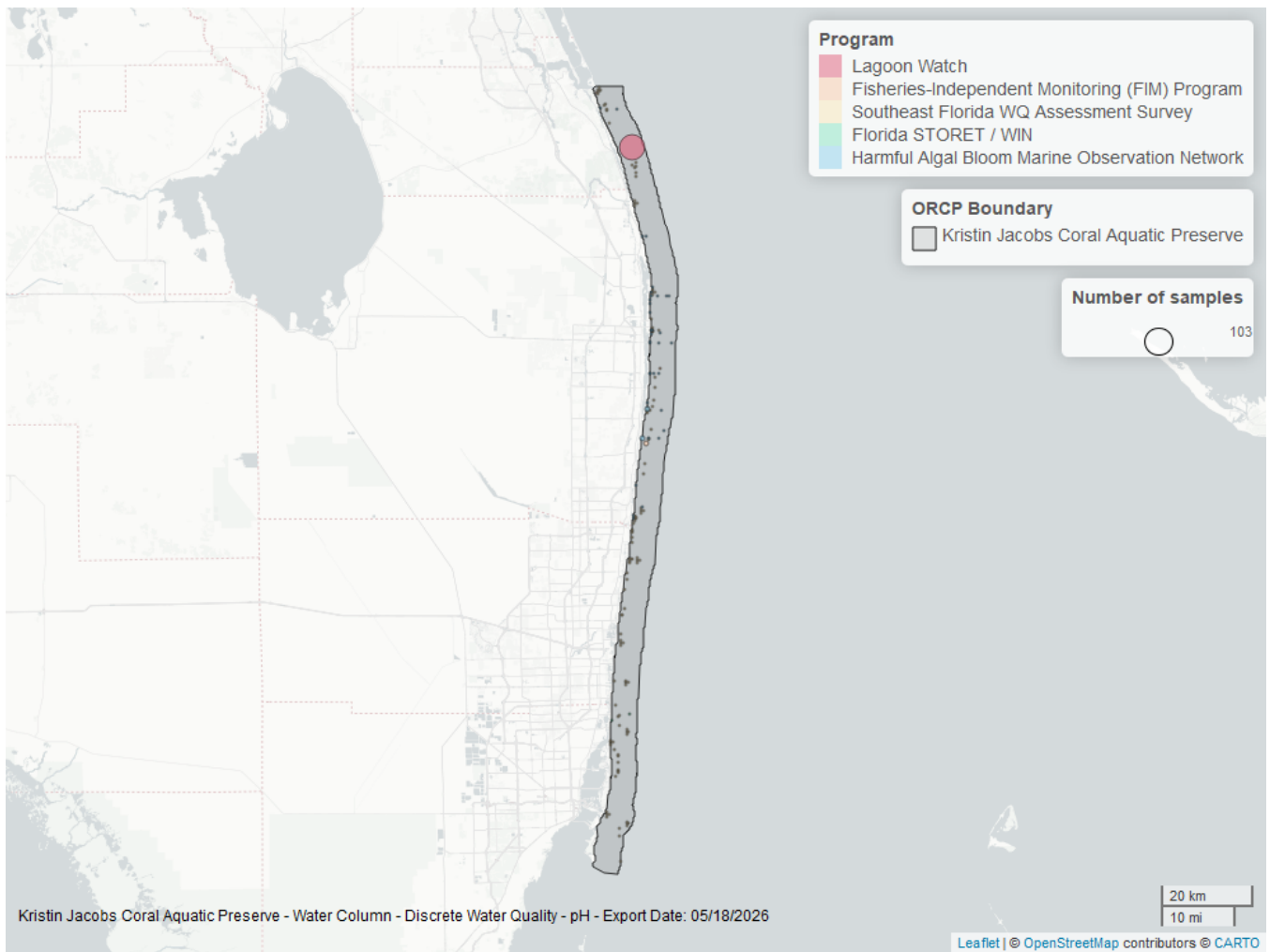


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5033	204	2025	2025
3001	103	1999	2003
95	62	1972	1972
69	7	1998	2024
5002	2	2011	2023

Program names:

[5033](#) - Southeast Florida Water Quality Assessment Survey⁶

3001 - Lagoon Watch³

95 - Harmful Algal Bloom Marine Observation Network⁴

69 - Fisheries-Independent Monitoring (FIM) Program⁵

5002 - Florida STORET / WIN¹

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

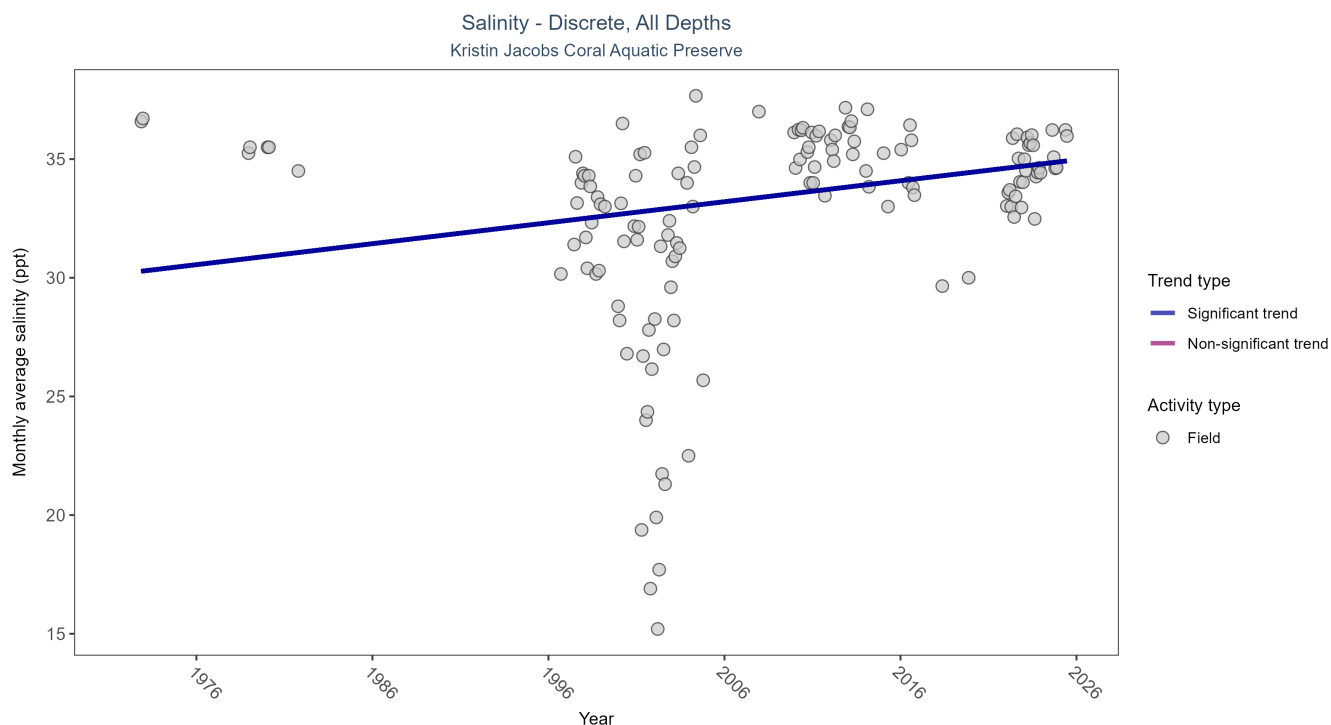


Figure 7: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	1895	29	1972 - 2025	36	0.2628	30.198	0.0884	0.0003

Monthly average salinity increased by 0.09 ppt per year.

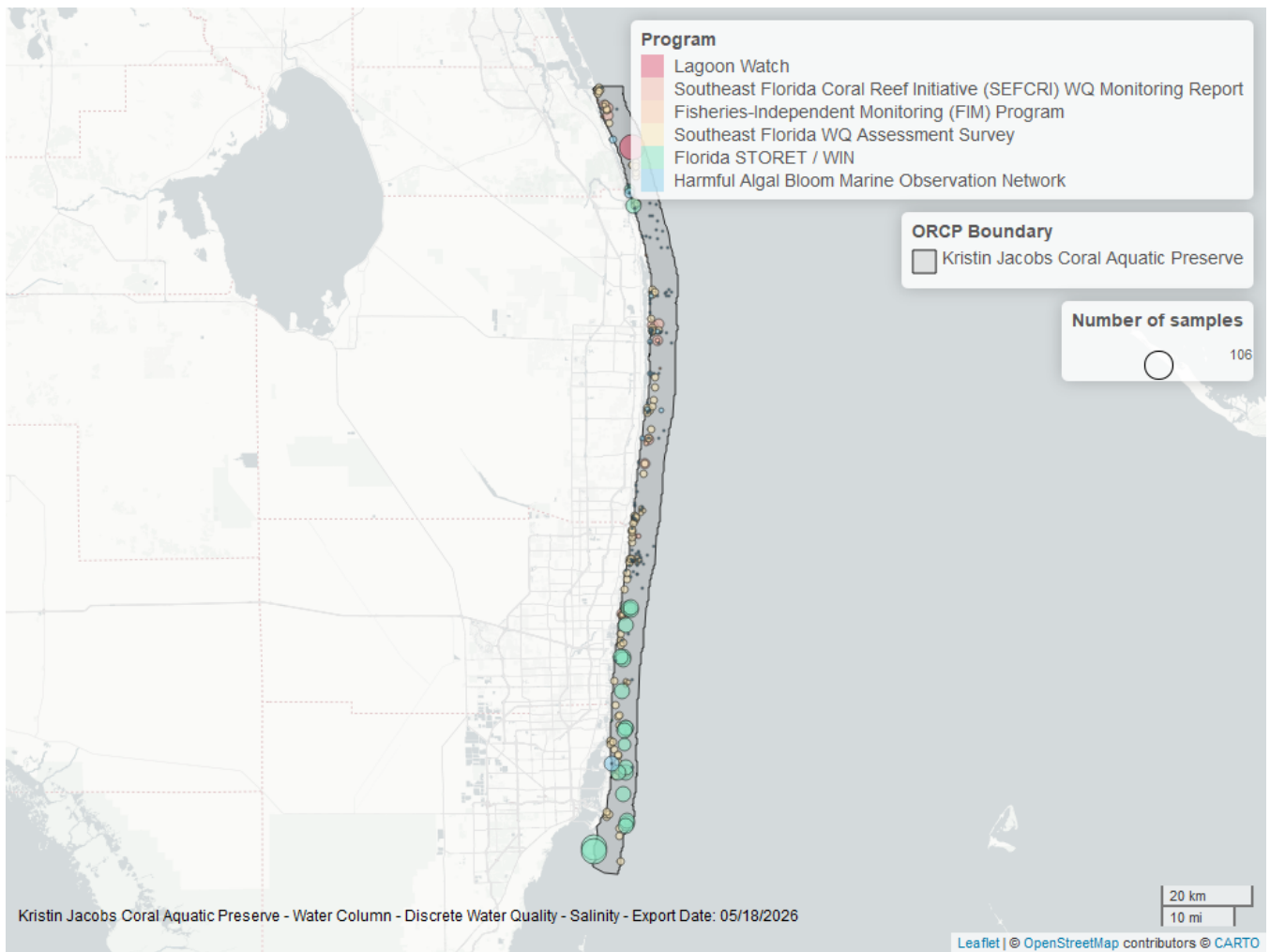


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5033	1042	2024	2025
5002	899	1996	2023
5058	266	2009	2011
95	199	1972	2018
3001	102	1999	2003
69	11	1997	2024

Program names:

[5033](#) - Southeast Florida Water Quality Assessment Survey⁶

[5002](#) - Florida STORET / WIN¹

[5058](#) - Southeast Florida Coral Reef Initiative (SEFCRI) Water Quality Monitoring Report²

95 - Harmful Algal Bloom Marine Observation Network⁴

3001 - Lagoon Watch³

69 - Fisheries-Independent Monitoring (FIM) Program⁵

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

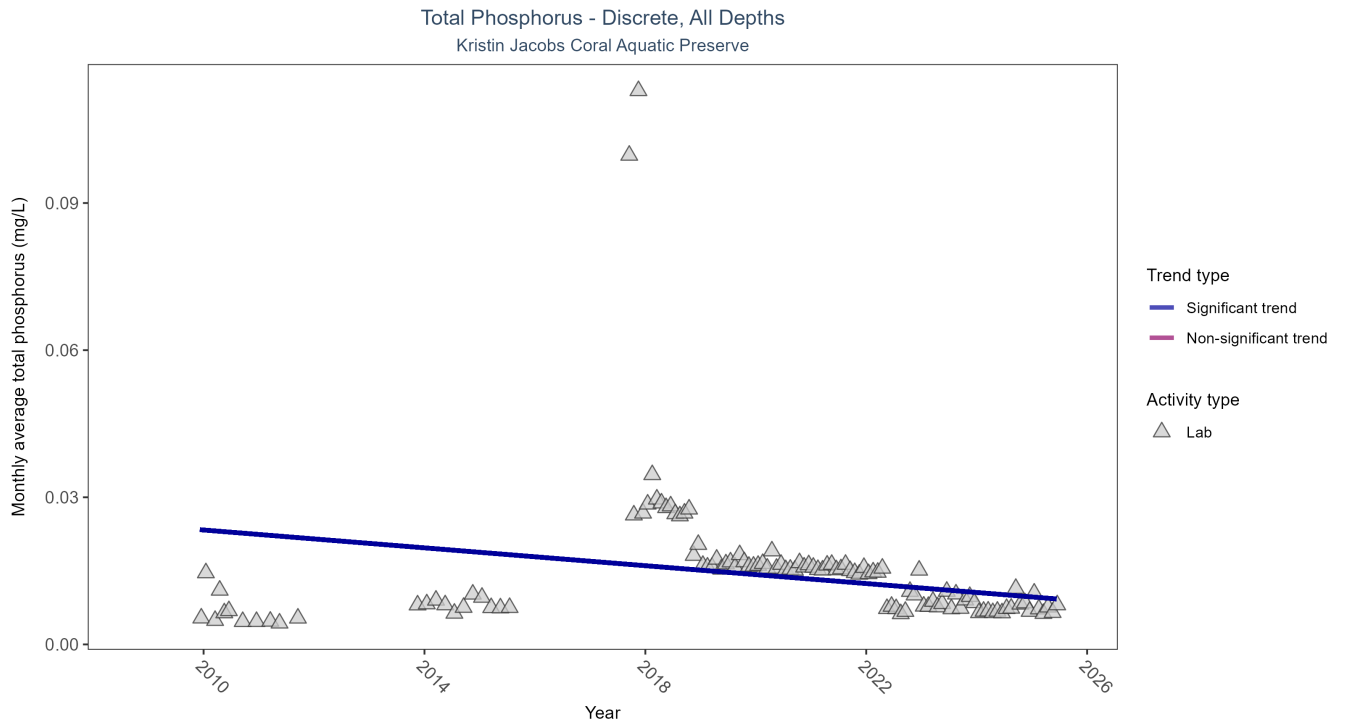


Figure 9: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	17904	15	2009 - 2025	0.014	-0.3362	0.0242	-0.0009	0.0001

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

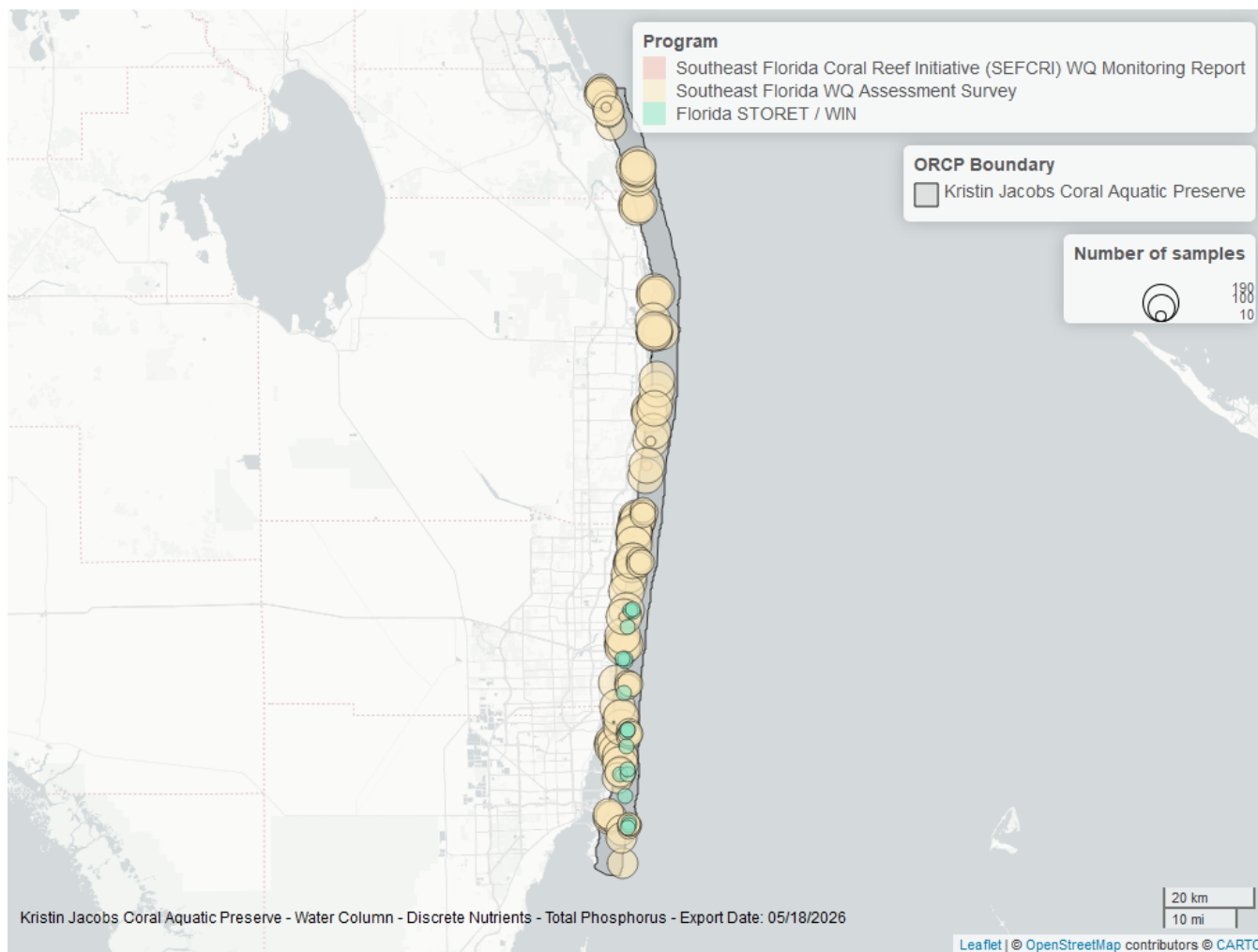


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5033	17934	2017	2025
5002	642	2013	2023
5058	268	2009	2011

Program names:

[5033](#) - Southeast Florida Water Quality Assessment Survey⁶

[5002](#) - Florida STORET / WIN¹

[5058](#) - Southeast Florida Coral Reef Initiative (SEFCRI) Water Quality Monitoring Report²

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

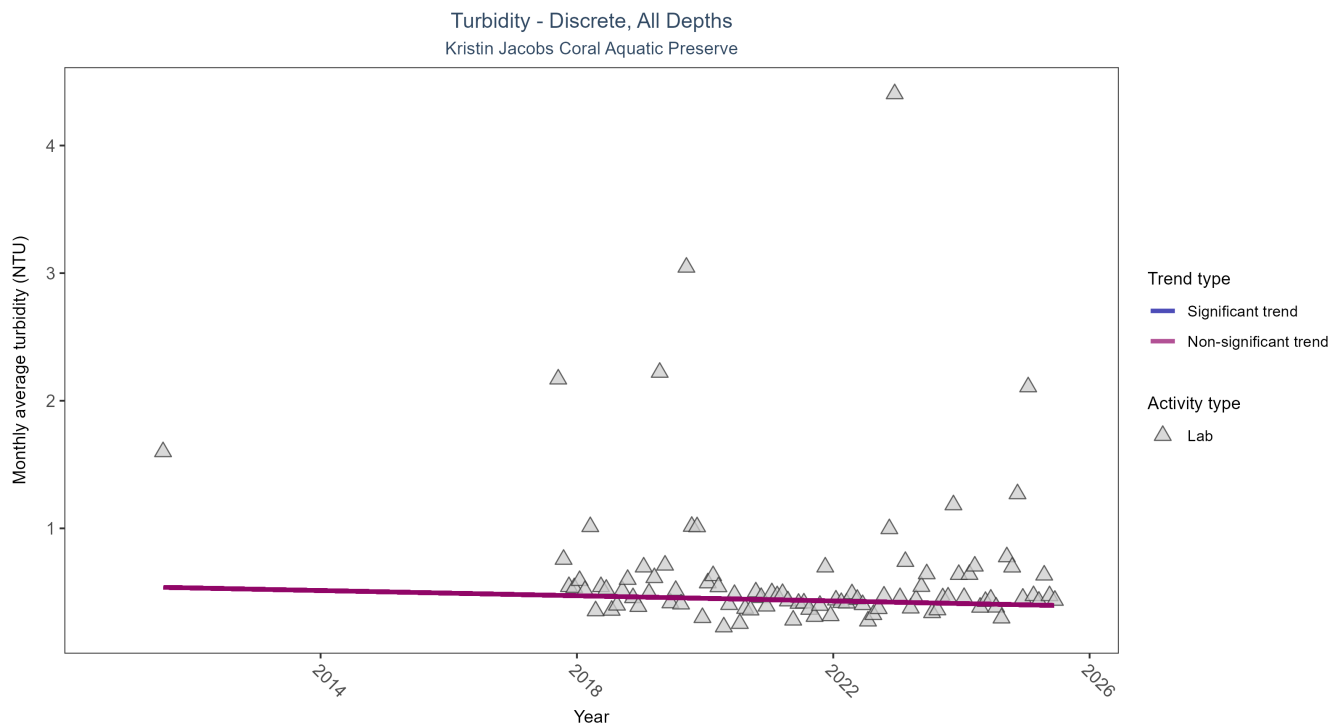


Figure 11: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	18027	10	2011 - 2025	0.3	-0.1138	0.5434	-0.0102	0.2184

Turbidity showed no detectable trend between 2011 and 2025.

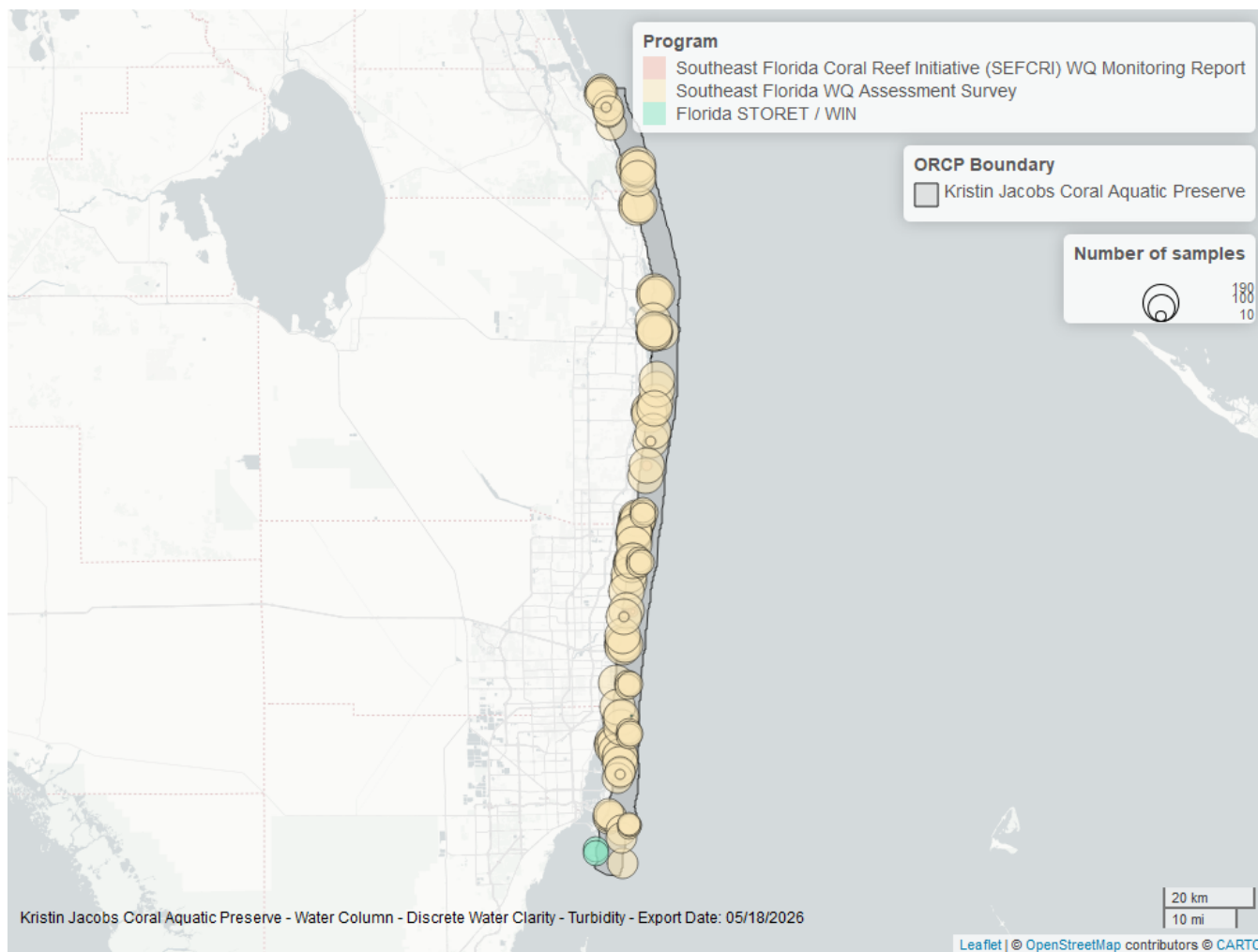


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5033	18051	2017	2025
5058	264	2009	2011
5002	210	2011	2023

Program names:

[5033](#) - Southeast Florida Water Quality Assessment Survey⁶

[5058](#) - Southeast Florida Coral Reef Initiative (SEFCRI) Water Quality Monitoring Report²

[5002](#) - Florida STORET / WIN¹

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

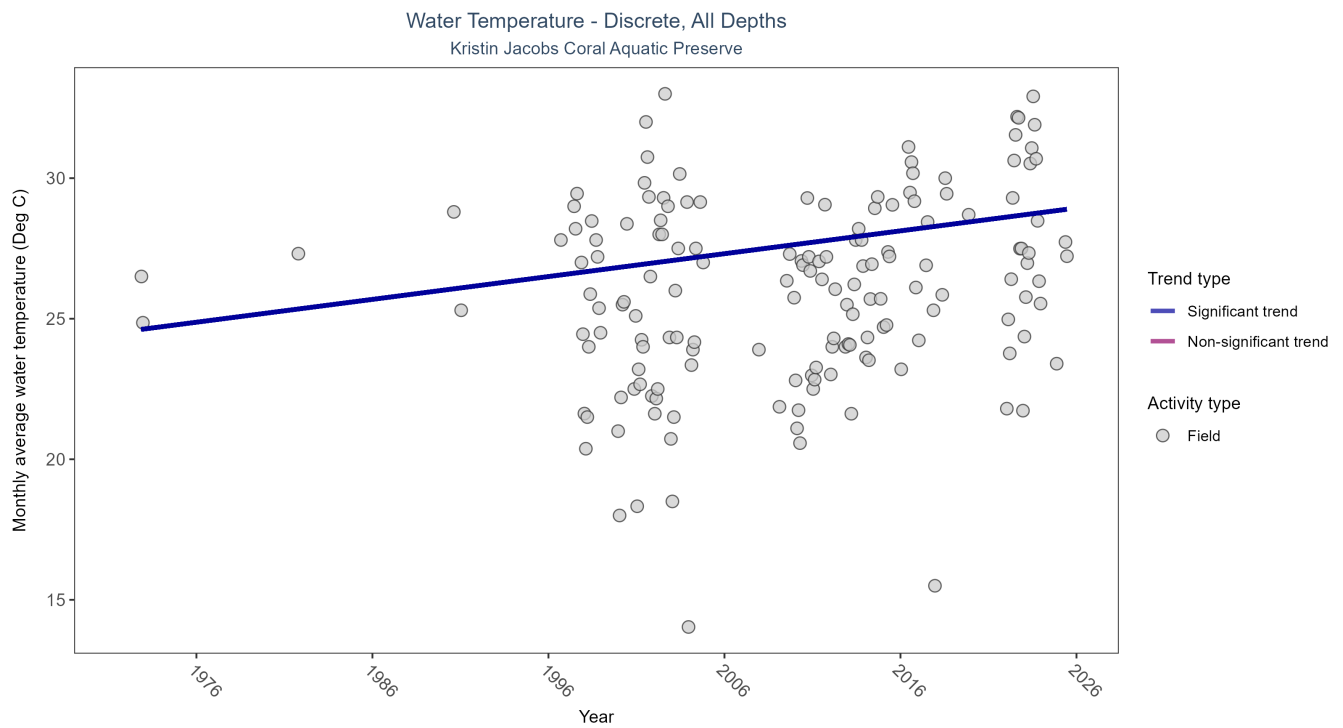


Figure 13: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	1928	29	1972 - 2025	27	0.2868	24.5503	0.0813	0

Monthly average water temperature increased by 0.08°C per year.

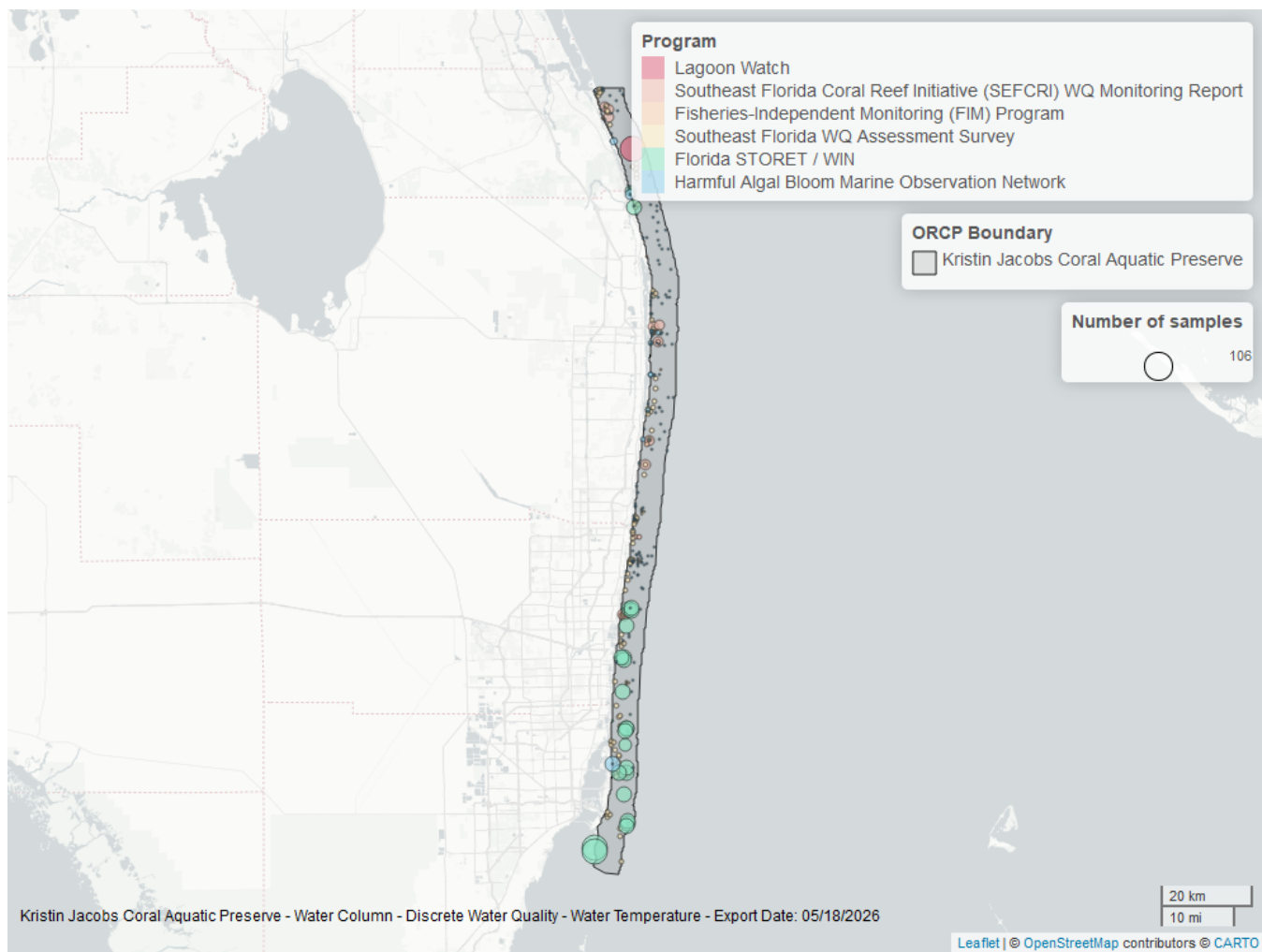


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	903	1996	2023
5033	408	2025	2025
5058	266	2009	2011
95	239	1972	2018
3001	104	1999	2003
69	11	1997	2024

Program names:

5002 - Florida STORET / WIN¹

5033 - Southeast Florida Water Quality Assessment Survey⁶

5058 - Southeast Florida Coral Reef Initiative (SEFCRI) Water Quality Monitoring Report²

95 - Harmful Algal Bloom Marine Observation Network⁴

3001 - Lagoon Watch³

69 - Fisheries-Independent Monitoring (FIM) Program⁵

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Kristin Jacobs Coral Aquatic Preserve

Table 20: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
5	LKWF1	42	TRUE	Water Temperature
986	1	11	TRUE	Water Temperature
986	2	11	TRUE	Water Temperature
986	3	11	TRUE	Water Temperature
986	4	11	TRUE	Water Temperature
986	5	11	TRUE	Water Temperature
986	6	11	TRUE	Water Temperature
986	84	17	TRUE	Water Temperature
986	85	17	TRUE	Water Temperature
986	86	17	TRUE	Water Temperature
986	87	17	TRUE	Water Temperature
986	88	17	TRUE	Water Temperature
986	89	17	TRUE	Water Temperature
986	90	17	TRUE	Water Temperature
986	91	17	TRUE	Water Temperature
986	92	17	TRUE	Water Temperature
986	93	17	TRUE	Water Temperature
986	94	17	TRUE	Water Temperature
986	95	17	TRUE	Water Temperature
986	96	6	TRUE	Water Temperature
986	97	14	TRUE	Water Temperature
986	98	14	TRUE	Water Temperature

Program names:

5 - National Data Buoy Center⁷

986 - Water Temperature on Coral Reefs in the Florida Keys⁸

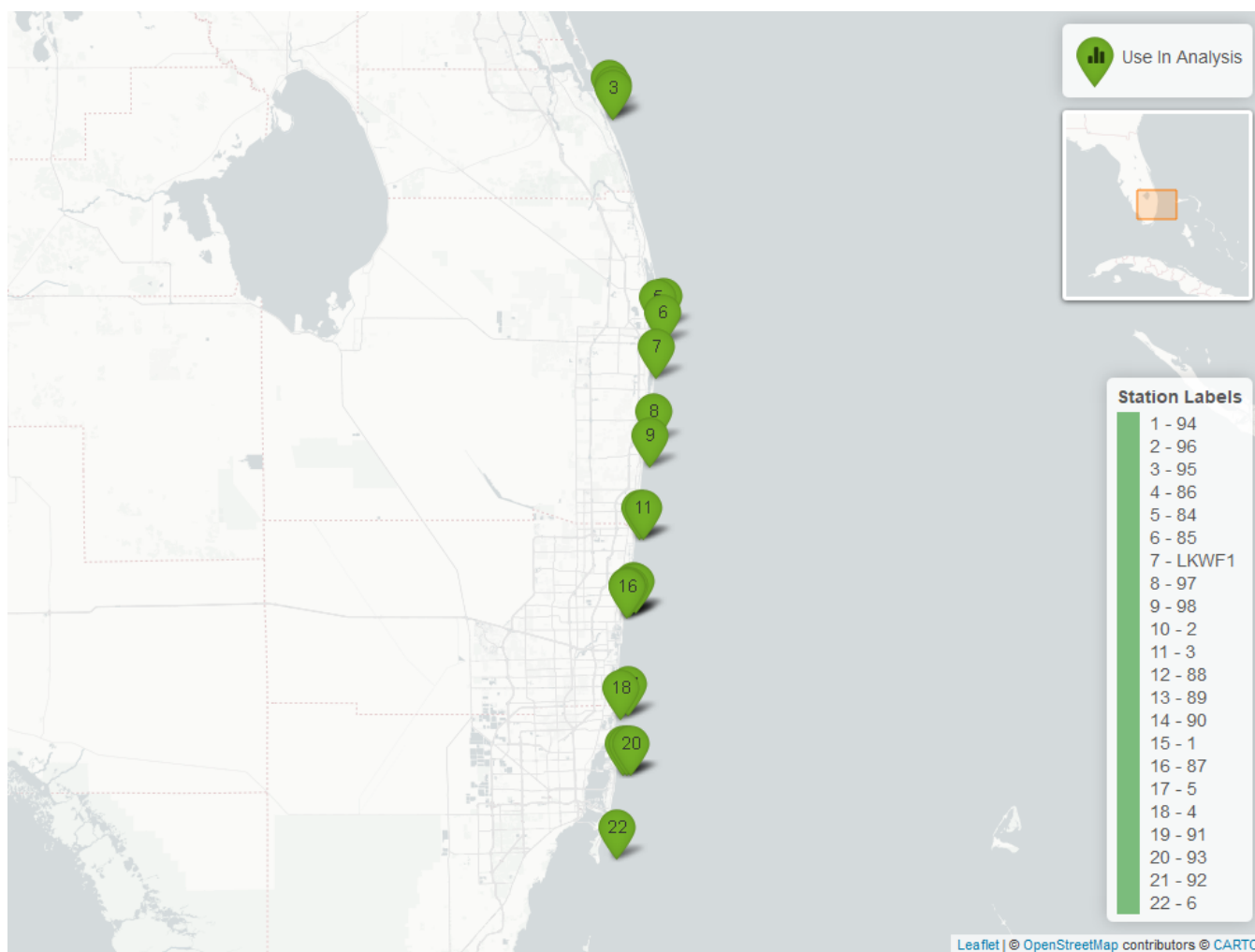


Figure 15: Map showing continuous water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

Water Temperature - Continuous - Program 5

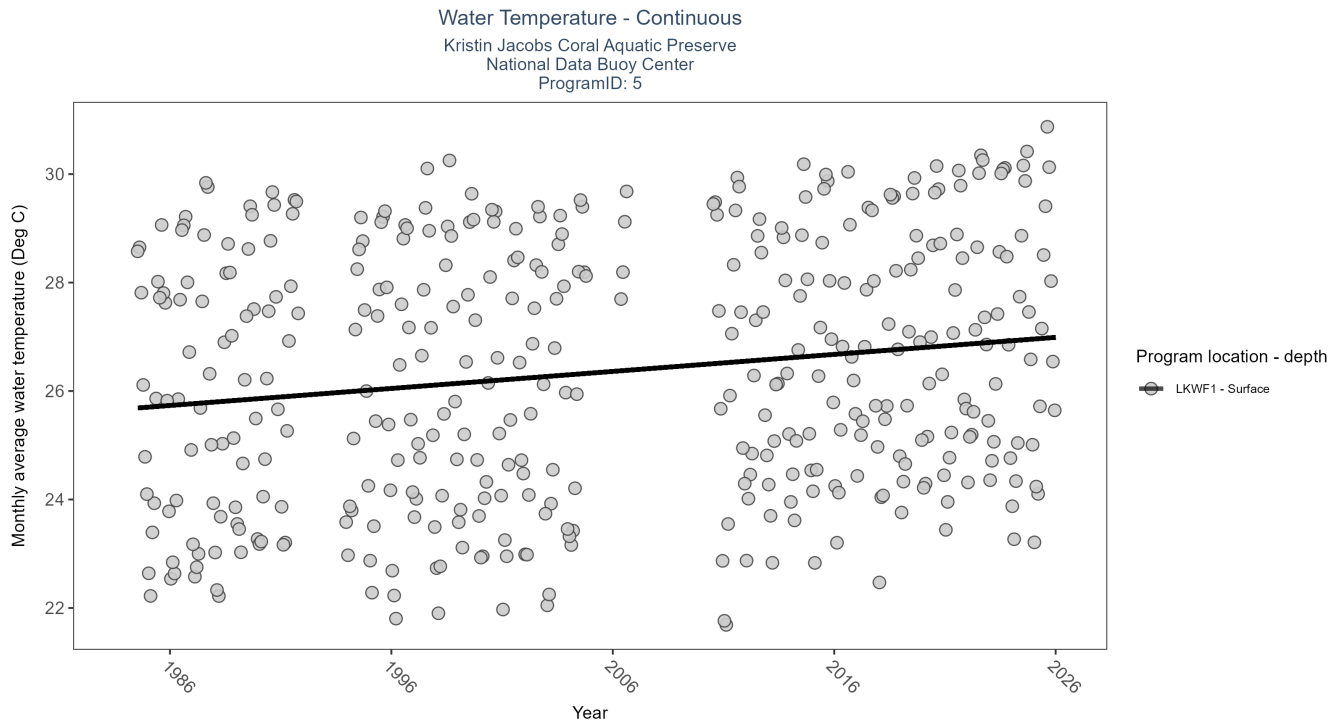


Figure 16: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 21: Seasonal Kendall-Tau Results for Water Temperature - Program 5

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
LKWF1	Significantly increasing trend	1378722	37	1984 - 2025	26.5	0.43	25.67	0.03	0

At twenty-one program locations, monthly average water temperature increased between 0.03 and 0.09°C per year. No detectable change in monthly average water temperature was observed at one location.

Water Temperature - Continuous - Program 986

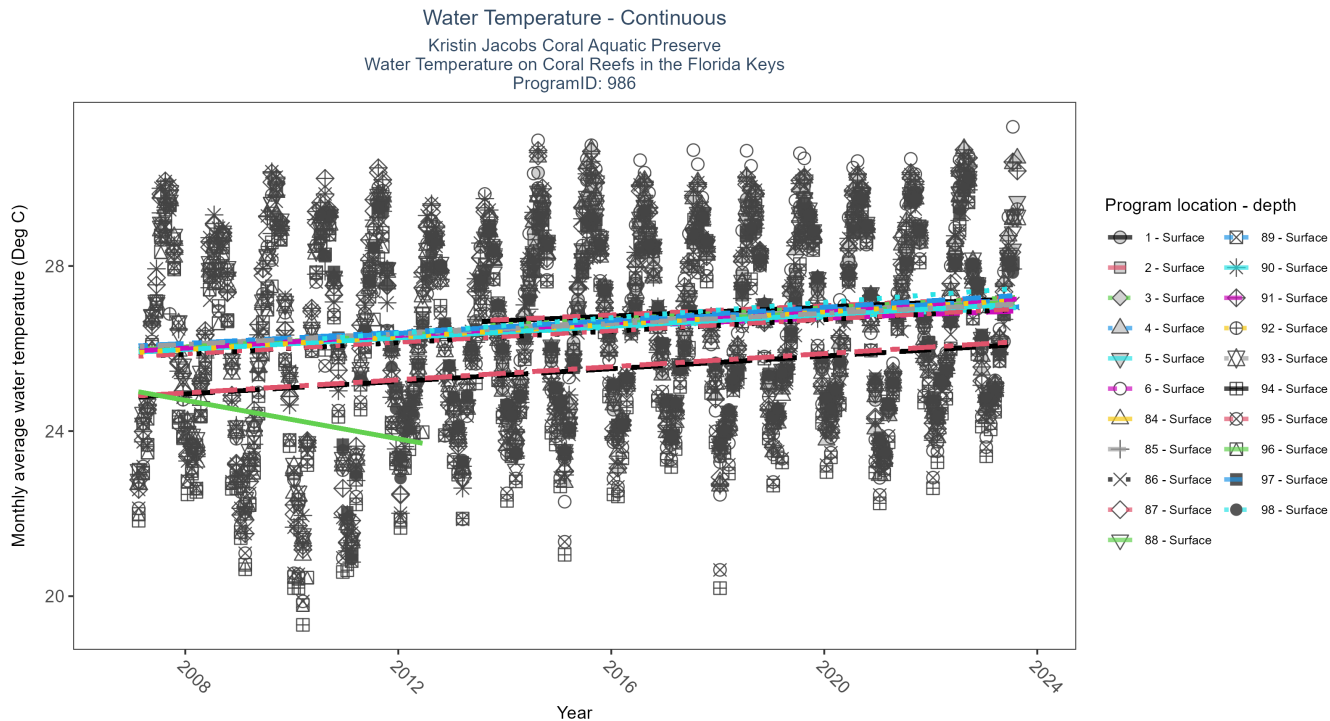


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 22: Seasonal Kendall-Tau Results for Water Temperature - Program 986

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
1	Significantly increasing trend	73700	11	2013 - 2023	26.50	0.23	26.62	0.06	0.00
2	Significantly increasing trend	68589	11	2013 - 2023	26.74	0.32	26.65	0.06	0.00
3	Significantly increasing trend	69054	11	2013 - 2023	26.69	0.32	26.51	0.07	0.00
4	Significantly increasing trend	77960	11	2013 - 2023	26.69	0.28	26.36	0.06	0.00
5	Significantly increasing trend	60729	11	2013 - 2023	26.69	0.28	26.33	0.06	0.00
84	Significantly increasing trend	117729	17	2007 - 2023	26.37	0.40	25.91	0.07	0.00
85	Significantly increasing trend	118414	17	2007 - 2023	26.29	0.38	26.03	0.07	0.00
86	Significantly increasing trend	113127	17	2007 - 2023	26.20	0.39	25.80	0.07	0.00
87	Significantly increasing trend	117324	17	2007 - 2023	26.50	0.37	25.81	0.07	0.00
88	Significantly increasing trend	123687	17	2007 - 2023	26.42	0.38	25.97	0.07	0.00
89	Significantly increasing trend	120476	17	2007 - 2023	26.32	0.38	26.06	0.07	0.00
90	Significantly increasing trend	103477	17	2007 - 2023	26.48	0.39	25.91	0.07	0.00
91	Significantly increasing trend	111048	17	2007 - 2023	26.60	0.38	25.93	0.08	0.00
92	Significantly increasing trend	120189	17	2007 - 2023	26.50	0.40	25.90	0.08	0.00
93	Significantly increasing trend	115688	17	2007 - 2023	26.56	0.38	26.04	0.06	0.00
94	Significantly increasing trend	98419	17	2007 - 2023	25.62	0.34	24.82	0.08	0.00
95	Significantly increasing trend	109401	17	2007 - 2023	25.62	0.36	24.85	0.08	0.00
96	No significant trend	25550	6	2007 - 2012	24.87	-0.25	24.98	-0.23	0.08
97	Significantly increasing trend	105832	14	2010 - 2023	26.50	0.42	26.21	0.08	0.00
98	Significantly increasing trend	94894	14	2010 - 2023	26.45	0.42	26.18	0.09	0.00
6	Significantly increasing trend	69753	11	2013 - 2023	26.84	0.19	26.41	0.05	0.01

At twenty-one program locations, monthly average water temperature increased between 0.03 and 0.09°C per year. No detectable change in monthly average water temperature was observed at one location.

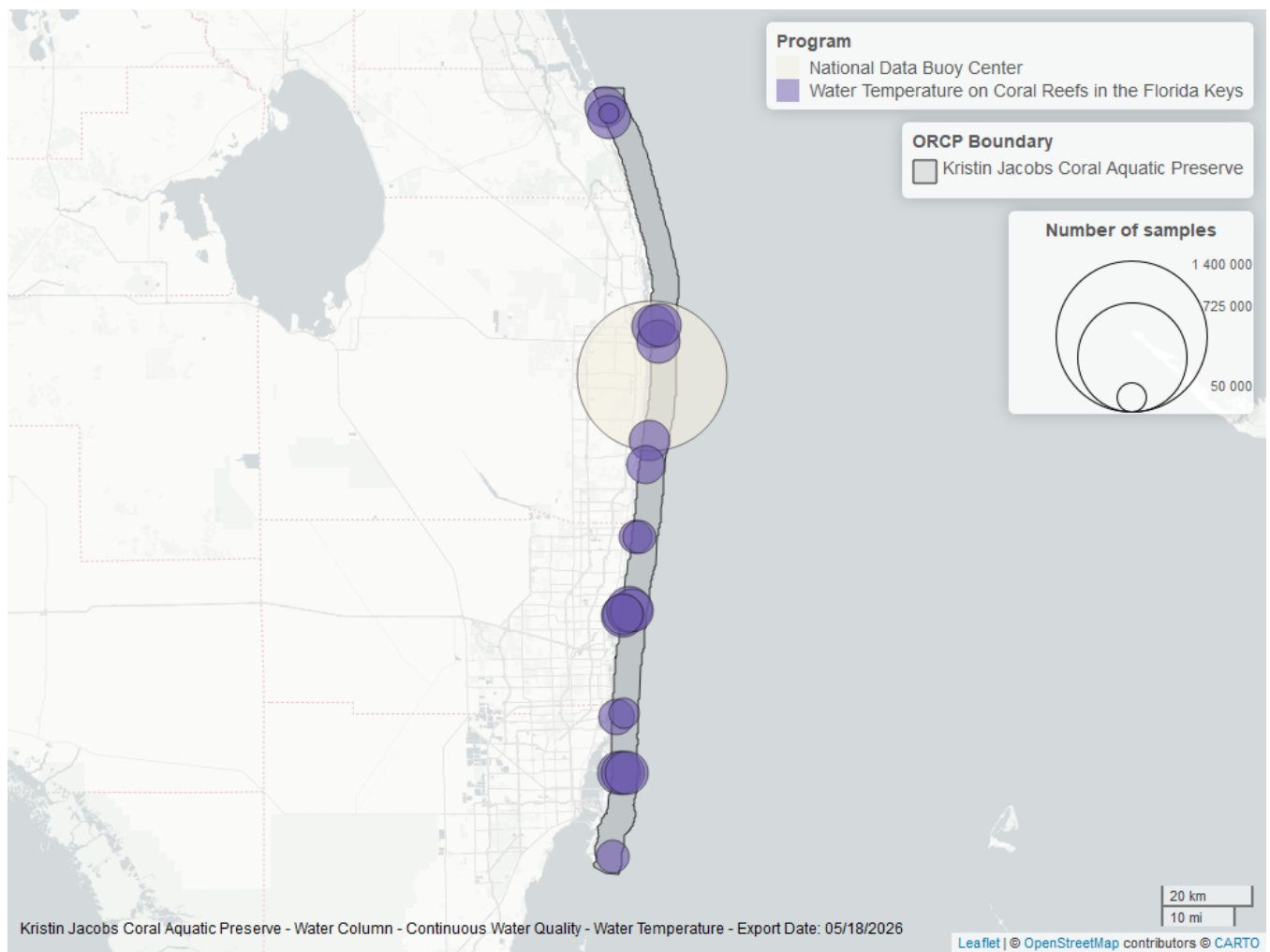


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Coral Reef

The data file used is: **All_CORAL_Parameters-2026-Jun-04.txt**

Percent Cover

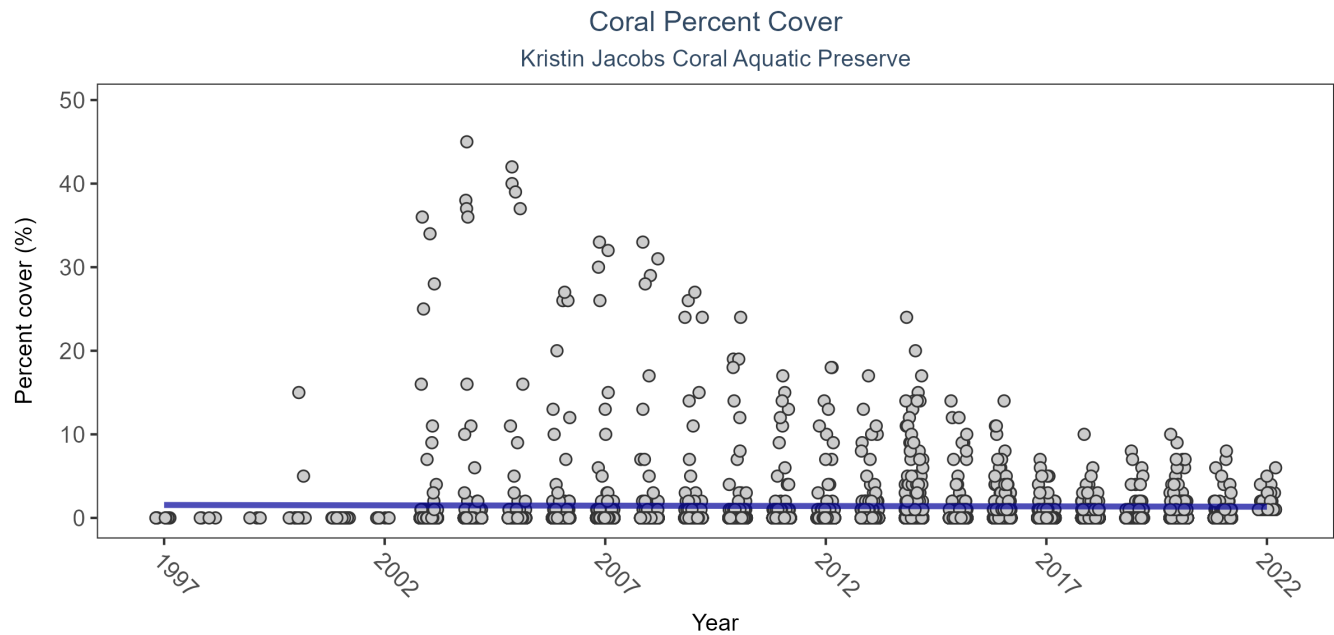


Figure 19: Scatter plot of live coral coverage over time as a percent of reef surface. Species groups include octocorals, milleporans, and scleractinians. If the time series included five or more years of observations, a significant (blue) or non-significant (magenta) trend line is also shown. Data points are jittered horizontally to reduce overlap.

Table 23: Coral Percent Cover

Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Decreasing trend	1997 - 2022	18.27128	-0.00837	0

Annual average percent cover decreased by -0.01% between 1997 and 2022.

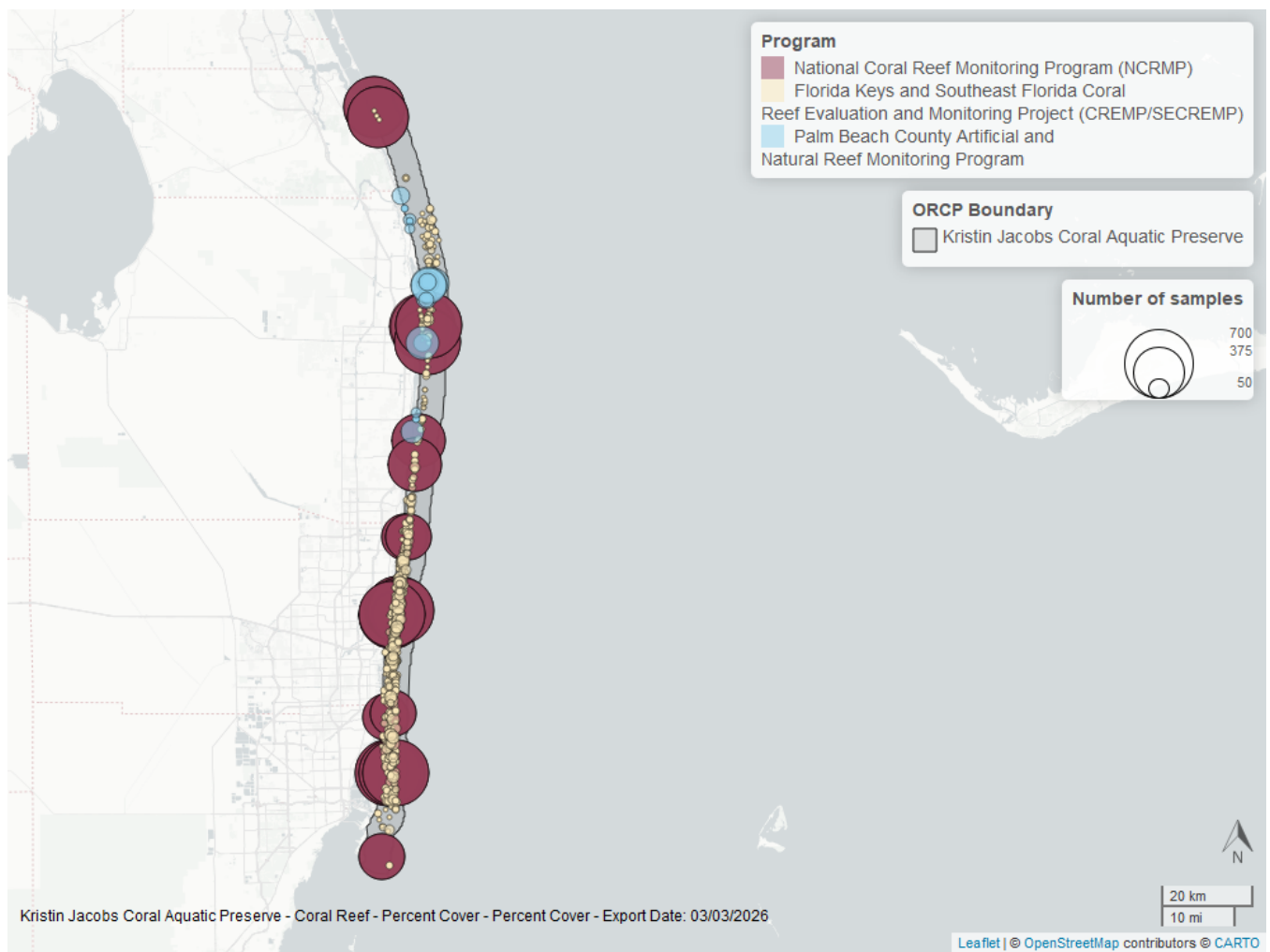


Figure 20: Map showing location of coral percent cover sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species Richness

Grazers and Reef-Dependent Species Richness
Kristin Jacobs Coral Aquatic Preserve

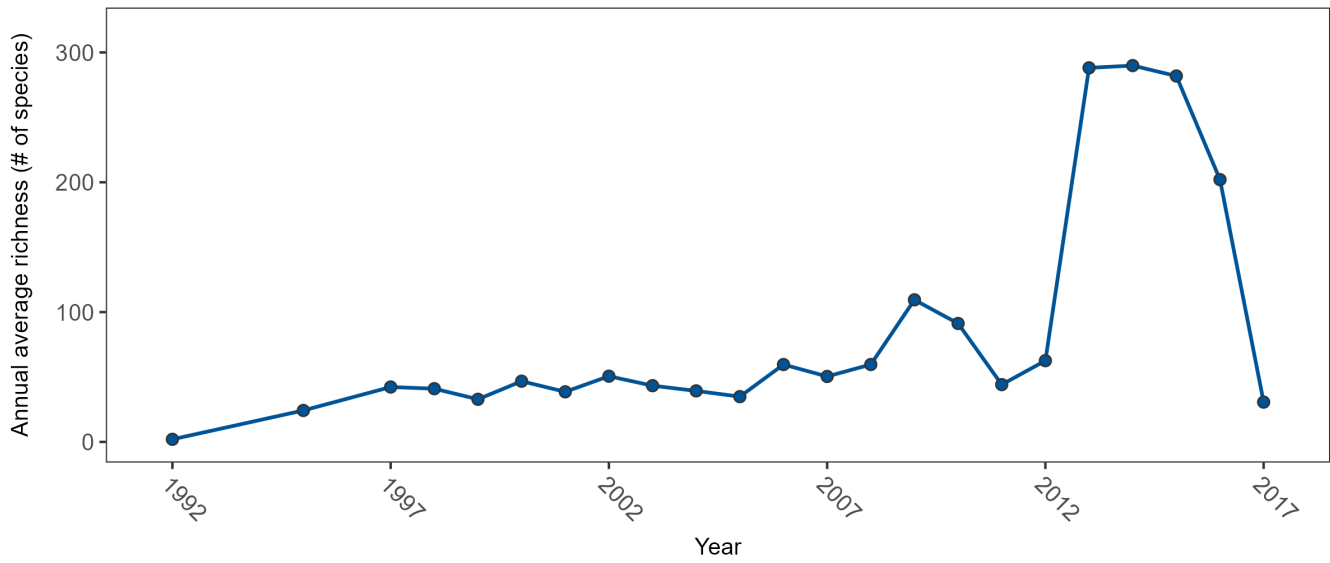


Figure 21: Line graph of annual average species richness of grazers and reef-dependent species over time. If the time series included more than one year of observations, a line connects the data points for visualization.

Table 24: Coral Species Richness

<i>No. of Samples</i>	<i>No. Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
3269	23	1992 - 2017	294	218.4475

The median annual number of taxa was 294 based on 3,269 observations collected between 1992 and 2017.

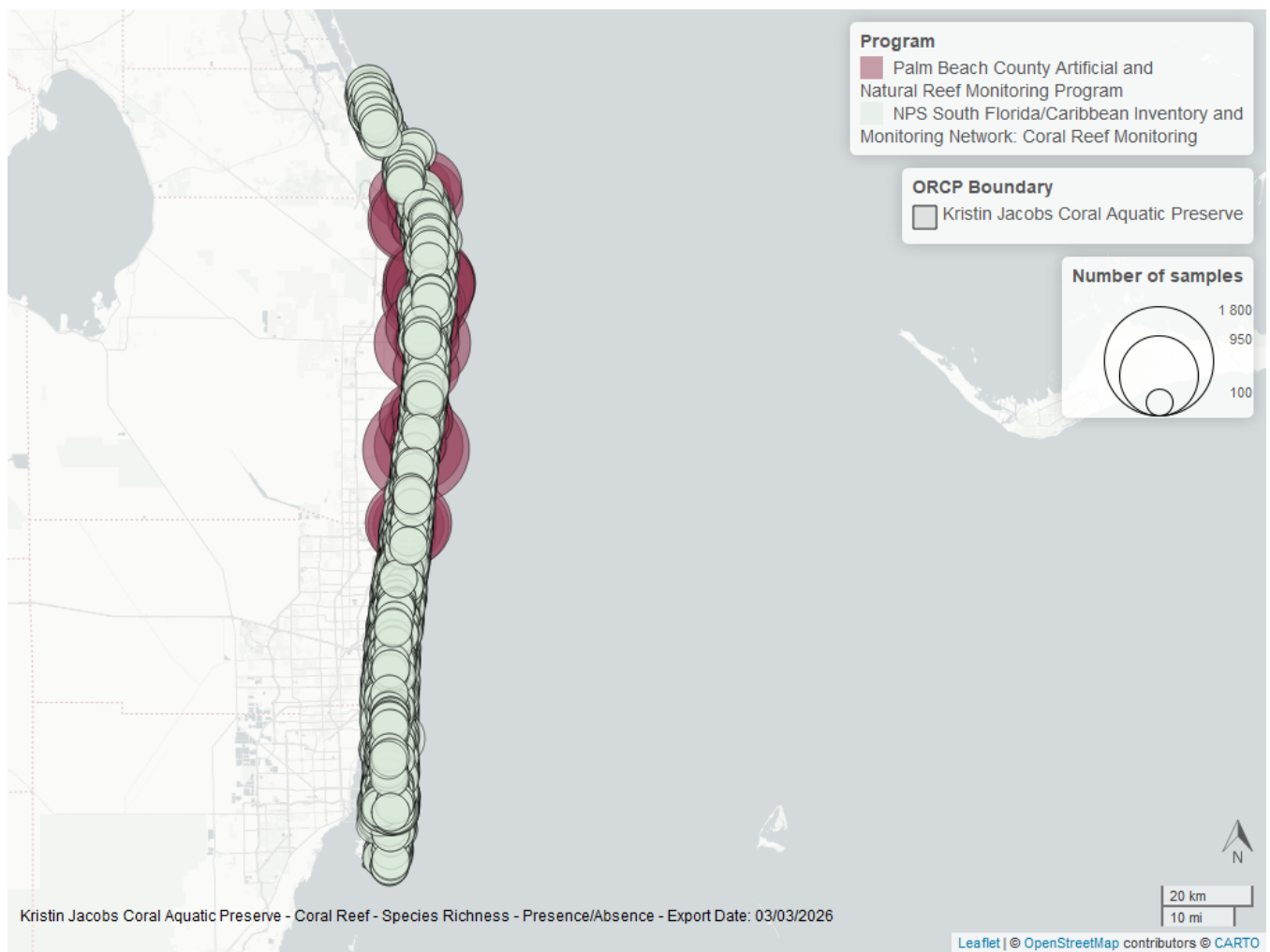


Figure 22: Map showing location of coral species richness sampling locations within the boundaries of *Kristin Jacobs Coral Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Abudefduf saxatilis ¹	Epinephelus itajara ¹	Opsanus tau ¹
Abudefduf taurus ¹	Epinephelus morio ¹	Orbicella annularis ²
Acanthemblemaria aspera ¹	Epinephelus striatus ¹	Orbicella faveolata ²
Acanthemblemaria chaplini ¹	Eques lanceolatus ¹	Orbicella franksi ²
Acanthemblemaria maria ¹	Equetus punctatus ¹	Orbicella sp.
Acanthemblemaria spinosa ¹	Erythropodium caribaeorum ²	Orthopristis chrysoptera ¹
Acanthocybium solandri	Eucidaris tribuloides	Other calcareous macroalgae
Acanthostracion polygonium ¹	Eucinostomus argenteus	Other fleshy macroalgae
Acanthostracion quadricornis ¹	Eucinostomus gula	Other red algae
Acanthurus bahianus ¹	Eucinostomus jonesii	Oxyurichthys stigmalophius ¹
Acanthurus chirurgus ¹	Eucinostomus melanopterus	Padina
Acanthurus coeruleus ¹	Eudistoma obscuratum	Pagrus pagrus ¹
Acanthurus sp. ¹	Eudistoma sp.	Palythoa spp.
Acanthurus tractus ¹	Eunicidae	Parablennius marmoreus ¹
Acropora cervicornis ²	Eusmilia fastigiata ²	Paraclinus marmoratus ¹
Aetobatus narinari	Euthynnus alletteratus	Paraclinus nigripinnis ¹
Agaricia agaricites ²	Favia fragum ²	Paralichthys albigutta ¹
Agaricia fragilis ²	Fistularia tabacaria ¹	Paranthias furcifer ¹
Agaricia grahamae	Fowlerichthys ocellatus ¹	Pareques acuminatus ¹
Agaricia humilis	Galeocерdo cuvier ¹	Pareques umbrosus ¹
Agaricia lamarcki ²	Gerreidae	Paroncheilus affinis ¹
Agaricia sp. ²	Gerres cinereus	Pempheris poeyi ¹
Agaricia spp.	Ginglymostoma cirratum	Pempheris schomburgkii ¹
Agelas clathrodes	Gnatholepis cauerensis ¹	Peyssonnelia
Agelas conifera	Gnatholepis thompsoni ¹	Phaeoptyx pigmentaria ¹
Aglaophenia latecarinata	Gobiidae ¹	Phaeoptyx xenus ¹
Ahlia egmontis	Gobioclinus bucciferus ¹	Phallusia nigra
Albula vulpes	Gobioclinus filamentosus ¹	Phorbas amaranthus
Alcyonacea sp. ²	Gobioclinus gobio ¹	Phragmatopoma caudata
Alectis ciliaris ¹	Gobioclinus kalisherae ¹	Plakortis angulospiculatus
Alphestes afer ¹	Gobiosoma sp. ¹	Platax orbicularis
Aluterus monoceros ¹	Gorgonia ²	Platybelone argalus
Aluterus schoepfii ¹	Gramma loreto ¹	Plectrypops retrospinis ¹
Aluterus scriptus ¹	Gramma melacara	Polycarpa spongiabilis
Aluterus sp. ¹	Green frondose algae	Pomacanthidae ¹
Amblycirrhitis pinos ¹	Gymnangium allmani	Pomacanthus arcuatus ¹
Amphimedon compressa	Gymnangium speciosum	Pomacanthus paru ¹
Anchoa lyolepis	Gymnothorax funebris ¹	Pomacentridae ¹
Ancylosetta dilecta	Gymnothorax miliaris ¹	Pomatomus saltatrix
Anisotremus surinamensis ¹	Gymnothorax moringa ¹	Porifera
Anisotremus virginicus ¹	Gymnothorax nigromarginatus ¹	Porifera spp.
Antennarius multiocellatus	Gymnothorax saxicola ¹	Porites astreoides ²
Anthozoa	Gymnothorax vicinus ¹	Porites cf. branneri
Antillogorgia sp. ²	Haemulidae ¹	Porites divaricata ²
Aplysina cauliformis	Haemulon album ¹	Porites furcata
Aplysina fistularis	Haemulon aurolineatum ¹	Porites porites ²
Apogon aurolineatus ¹	Haemulon carbonarium ¹	Porites sp.
Apogon binotatus ¹	Haemulon flavolineatum ¹	Porites spp. ²
Apogon lachneri ¹	Haemulon macrostomum ¹	Porolithon pachydermum
Apogon maculatus ¹	Haemulon melanurum ¹	Priacanthus arenatus ¹
Apogon phenax ¹	Haemulon parra ¹	Priolepis hipoliti ¹
Apogon planifrons ¹	Haemulon plumierii ¹	Prionotus ophryas ¹
Apogon pseudomaculatus ¹	Haemulon sciurus ¹	Prionotus rubio ¹
Apogon quadrisquamatus ¹	Haemulon sp. ¹	Prionotus tribulus ¹

Apogon robinsoni ¹	Haemulon striatum ¹	Pristipomoides aquilonaris ¹
Apogon townsendi ¹	Haemulon vittatum ¹	Pristis pectinata
Archosargus probatocephalus ¹	Halichoeres bivittatus ¹	Prognathodes aculeatus ¹
Archosargus rhomboidalis ¹	Halichoeres caudalis ¹	Prognathodes aya ¹
Ascidacea	Halichoeres cyanocephalus ¹	Pseudobatos lentiginosus
Asteroidea	Halichoeres garnoti ¹	Pseudodiploria clivosa ²
Astrapogon punctulatus ¹	Halichoeres maculipinna ¹	Pseudodiploria sp.
Astrapogon sp. ¹	Halichoeres pictus ¹	Pseudodiploria strigosa ²
Astrapogon stellatus ¹	Halichoeres poeyi ¹	Pseudoplexaura ²
Astroscopus guttatus	Halichoeres radiatus ¹	Pseudupeneus maculatus ¹
Astroscopus ygraecum	Halimeda	Ptereleotris calliura
Atherinomorus stipes	Halimeda goreau	Ptereleotris helenae
Aulostomus maculatus ¹	Halimeda opuntia	Pterogorgia ²
Azurina cyanea ¹	Halimeda spp.	Pterois miles ¹
Balistes capriscus ¹	Halisarca sp.	Pterois sp. ¹
Balistes sp. ¹	Halopteris carinata	Pterois volitans ¹
Balistes vetula ¹	Harengula humeralis	Rachycentron canadum
Balistidae ¹	Harengula jaguana	Raja eglanteria
Bare substrate	Helioseris cucullata	Ramirostia spp.
Bathytoshia centroura	Hemiblemnia simula ¹	Razorfish sp. ¹
Blenniidae ¹	Hemiramphus balao	Remora remora
Blenniidae sp. ¹	Hemiramphus brasiliensis	Rhinoptera bonasus
Bodianus pulchellus ¹	Hermodice carunculata	Rhomboplites aurorubens ¹
Bodianus rufus ¹	Heteroconger longissimus	Rubble
Bollmannia boqueronensis ¹	Heteroconger luteolus	Rypticus bistrispinus ¹
Bothus lunatus ¹	Heteropriacanthus cruentatus ¹	Rypticus maculatus ¹
Bothus ocellatus ¹	Hippocampus erectus ¹	Rypticus saponaceus ¹
Brachygenys chrysargyreum ¹	Hippocampus reidi ¹	Rypticus subbifrenatus ¹
Brockius nigricinctus ¹	Hippopodina feegeensis	Sabellidae
Brotula barbata	Holacanthus bermudensis ¹	Sand-sand
Bryozoa	Holacanthus ciliaris ¹	Sand on hard-bottom
Calamus bajonado ¹	Holacanthus spp. ¹	Sardinella aurita
Calamus calamus ¹	Holacanthus tricolor ¹	Sargassum sp.
Calamus leucosteus ¹	Holocentrus adscensionis ¹	Sargocentron bullisi ¹
Calamus nodosus ¹	Holocentrus rufus ¹	Sargocentron coruscum ¹
Calamus penna ¹	Hypanus americanus	Sargocentron vexillarium ¹
Calamus proridens ¹	Hypleurochilus bermudensis ¹	Scaridae ¹
Callionymus bairdi ¹	Hypoatherina harringtonensis	Scartella cristata ¹
Callyspongia (Cladochalina) aculeata	Hypoplectrus chlorurus ¹	Scarus coelestinus ¹
Callyspongia (Cladochalina) plicifera	Hypoplectrus gemma ¹	Scarus coerulesus ¹
Cantherhines macrocerus ¹	Hypoplectrus gummigutta ¹	Scarus guacamaia ¹
Cantherhines pullus ¹	Hypoplectrus guttavarius ¹	Scarus iseri ¹
Canthidermis sufflamen ¹	Hypoplectrus hybrid ¹	Scarus sp. ¹
Canthigaster rostrata ¹	Hypoplectrus indigo ¹	Scarus taeniopterus ¹
Carangidae ¹	Hypoplectrus nigricans ¹	Scarus vetula ¹
Caranx bartholomaei ¹	Hypoplectrus puella ¹	Schizoporella
Caranx crysos ¹	Hypoplectrus randallorum ¹	Schultzea beta ¹
Caranx hippos ¹	Hypoplectrus sp. ¹	Sciaenidae ¹
Caranx latus ¹	Hypoplectrus tann ¹	Scianid sp. ¹
Caranx lugubris ¹	Hypoplectrus unicolor ¹	Scleractinia ²
Caranx ruber ¹	Hyporthodus flavolimbatus ¹	Scolymia cubensis
Caranx sp. ¹	Hyporthodus niveatus ¹	Scolymia sp. ²
Carcharhinus falciformis	Iciligorgia schrammi ²	Scolymia spp. ²
Carcharhinus leucas	Iotrochota birotulata	Scomberomorus cavalla
Carcharhinus limbatus	Ircinia strobilina	Scomberomorus maculatus
Carcharhinus obscurus	Isophyllia rigida	Scomberomorus regalis
Carcharhinus perezii	Isophyllia sinuosa ²	Scopalina ruetzleri

<i>Carcharhinus plumbeus</i>	<i>Isophyllia</i> sp.	<i>Scorpaena brasiliensis</i>
<i>Carijoa riisei</i> ²	<i>Istiophorus platypterus</i>	<i>Scorpaena plumieri</i> ¹
<i>Caulerpa</i>	<i>Jenkinsia</i> sp.	<i>Scorpaenodes caribbaeus</i> ¹
<i>Caulerpa racemosa</i>	<i>Kyphosus sectatrix</i> ¹	<i>Selachii</i>
<i>Centropomus undecimalis</i>	<i>Kyphosus vaigiensis</i> ¹	<i>Selar crumenophthalmus</i> ¹
<i>Centropristis ocyurus</i> ¹	<i>Labridae</i> ¹	<i>Selene vomer</i> ¹
<i>Centropristis philadelphica</i> ¹	<i>Labrisomidae</i> sp. ¹	<i>Seriola dumerili</i> ¹
<i>Centropristis striata</i> ¹	<i>Labrisomus nuchipinnis</i> ¹	<i>Seriola rivoliana</i> ¹
<i>Centropyge argi</i> ¹	<i>Labrisomus</i> spp. ¹	<i>Seriola</i> sp. ¹
<i>Cephalopholis cruentata</i> ¹	<i>Lachnolaimus maximus</i> ¹	<i>Seriola zonata</i> ¹
<i>Cephalopholis fulva</i> ¹	<i>Lactophrys bicaudalis</i> ¹	<i>Serranid</i> sp. ¹
<i>Ceramium</i> sp.	<i>Lactophrys trigonus</i> ¹	<i>Serranidae</i> ¹
<i>Chaenopsis limbaughii</i> ¹	<i>Lactophrys triqueter</i> ¹	<i>Serranus annularis</i> ¹
<i>Chaetodipterus faber</i>	<i>Lagodon rhomboides</i> ¹	<i>Serranus baldwini</i> ¹
<i>Chaetodon capistratus</i> ¹	<i>Leiostomus xanthurus</i> ¹	<i>Serranus chionaraia</i> ¹
<i>Chaetodon ocellatus</i> ¹	<i>Leptogorgia hebes</i> ²	<i>Serranus phoebe</i> ¹
<i>Chaetodon sedentarius</i> ¹	<i>Liopropoma eukrines</i> ¹	<i>Serranus subligarius</i> ¹
<i>Chaetodon striatus</i> ¹	<i>Liopropoma mowbrayi</i> ¹	<i>Serranus tabacarius</i> ¹
<i>Chaetodontidae</i> ¹	<i>Liopropoma rubre</i> ¹	<i>Serranus tigrinus</i> ¹
<i>Chilomycterus antennatus</i> ¹	<i>Lobophora</i> spp.	<i>Serranus tortugarum</i> ¹
<i>Chilomycterus antillarum</i>	<i>Lutjanidae</i> ¹	<i>Sertularella diaphana</i>
<i>Chilomycterus reticulatus</i> ¹	<i>Lutjanus analis</i> ¹	<i>Siderastrea radians</i> ²
<i>Chilomycterus schoepfii</i> ¹	<i>Lutjanus apodus</i> ¹	<i>Siderastrea siderea</i> ²
<i>Chloroscombrus chrysurus</i> ¹	<i>Lutjanus buccanella</i> ¹	<i>Siderastrea</i> sp.
<i>Chriodorus atherinoides</i>	<i>Lutjanus campechanus</i> ¹	<i>Snapper</i> sp. ¹
<i>Chromis enchrysurus</i> ¹	<i>Lutjanus cyanopterus</i> ¹	<i>Solenastrea bournoni</i> ²
<i>Chromis insolata</i> ¹	<i>Lutjanus griseus</i> ¹	<i>Solenastrea hyades</i>
<i>Chromis multilineata</i> ¹	<i>Lutjanus jocu</i> ¹	<i>Sparidae</i> ¹
<i>Chromis scotti</i> ¹	<i>Lutjanus mahogoni</i> ¹	<i>Sparidae</i> sp. ¹
<i>Cirripecta</i>	<i>Lutjanus synagris</i> ¹	<i>Sparisoma atomarium</i> ¹
<i>Cladocora</i> ²	<i>Macroalgae</i>	<i>Sparisoma aurofrenatum</i> ¹
<i>Clathria (Thalysias) venosa</i>	<i>Macrorhynchia clarkei</i>	<i>Sparisoma chrysotermum</i> ¹
<i>Clathria</i> sp.	<i>Madracis auretenra</i> ²	<i>Sparisoma radians</i> ¹
<i>Clavelina picta</i>	<i>Madracis decactis</i> ²	<i>Sparisoma rubripinne</i> ¹
<i>Clavelina</i> sp.	<i>Madracis formosa</i>	<i>Sparisoma</i> sp. ¹
<i>Clepticus parrae</i> ¹	<i>Madracis senaria</i>	<i>Sparisoma viride</i> ¹
<i>Cliona caribbaea</i>	<i>Madracis</i> sp. ²	<i>Sphoeroides</i> ¹
<i>Cliona</i> sp.	<i>Madracis</i> spp. ²	<i>Sphoeroides nephelus</i> ¹
<i>Cliona</i> spp.	<i>Malacanthus plumieri</i>	<i>Sphoeroides spengleri</i> ¹
<i>Cliona varians</i>	<i>Malacoctenus aurolineatus</i> ¹	<i>Sphoeroides testudineus</i> ¹
<i>Clupeidae</i>	<i>Malacoctenus gilli</i> ¹	<i>Sphyraena barracuda</i> ¹
<i>Cnidaria</i>	<i>Malacoctenus macropus</i> ¹	<i>Sphyraena guachancho</i>
<i>Codium isthmocladum</i>	<i>Malacoctenus triangulatus</i> ¹	<i>Sphyraena picudilla</i>
<i>Codium</i> sp.	<i>Malacoctenus versicolor</i> ¹	<i>Sphyrna lewini</i>
<i>Colpophyllia natans</i> ²	<i>Manicina areolata</i>	<i>Sphyrna mokarran</i>
<i>Conger oceanicus</i>	<i>Meandrina danae</i>	<i>Sphyrna tiburo</i>
<i>Conger triporiceps</i>	<i>Meandrina jacksoni</i>	<i>Sphyrna zygaena</i>
<i>Corvula sanctaeluciae</i> ¹	<i>Meandrina meandrites</i> ²	<i>Spirastrella coccinea</i>
<i>Coryphopterus dicrus</i> ¹	<i>Meandrina</i> sp.	<i>Spondylus americanus</i>
<i>Coryphopterus eidolon</i> ¹	<i>Megalops atlanticus</i>	<i>Squirrelfish</i> sp. ¹
<i>Coryphopterus glaucofraenum</i> ¹	<i>Melichthys niger</i> ¹	<i>Stegastes adustus</i> ¹
<i>Coryphopterus lipernes</i> ¹	<i>Menidia</i> sp.	<i>Stegastes diencaeus</i> ¹
<i>Coryphopterus personatus</i> ¹	<i>Micrognathus crinitus</i> ¹	<i>Stegastes leucostictus</i> ¹
<i>Coryphopterus punctipictophorus</i> ¹	<i>Microgobius carri</i> ¹	<i>Stegastes partitus</i> ¹
<i>Coryphopterus</i> sp. ¹	<i>Microgobius microlepis</i> ¹	<i>Stegastes planifrons</i> ¹
<i>Cosmocampus albirostris</i> ¹	<i>Microspathodon chrysurus</i> ¹	<i>Stegastes</i> sp. ¹
<i>Crustacea</i>	<i>Millepora alcornis</i> ²	<i>Stegastes variabilis</i> ¹

Crustose coralline algae	Millepora complanata ²	Stephanocoenia intersepta ²
Cryptotomus roseus ¹	Millepora sp. ²	Stephanolepis hispida ¹
Ctenogobius saepepallens ¹	Millepora spp. ²	Stephanolepis setifer ¹
Cyanobacteria	Mobula birostris	Strongylacidon sp.
Cyanophyta spp.	Mollusca	Strongylura marina
Cynoscion nebulosus ¹	Monacanthidae ¹	Strongylura notata ¹
Dactylopterus volitans	Monacanthus ciliatus ¹	Strongylura timucu
Decapterus macarellus ¹	Monacanthus tuckeri ¹	Stygnobrotula latebricola
Decapterus punctatus ¹	Monanchora arbuscula	Substrate
Decapterus sp. ¹	Montastraea cavernosa ²	Swiftia exserta ²
Dendrogyra cylindrus ²	Mugil cephalus	Syacium micrurum ¹
Dentitheca dendritica	Mullidae ¹	Symplegma viride
Dermatolepis inermis ¹	Mulloidichthys martinicus ¹	Syngnathus louisianae ¹
Desmapsamma anchorata	Muraena retifera ¹	Syngnathus scovelli ¹
Diadema antillarum	Muricea ²	Syngnathus sp. ¹
Dichocoenia stokesii ²	Mussa angulosa ²	Synodus foetens ¹
Dictyota	Mycetophyllia aliciae ²	Synodus intermedius ¹
Dictyota spp.	Mycetophyllia ferox	Synodus saurus ¹
Didemnidae	Mycetophyllia lamarckiana	Synodus synodus ¹
Diodogorgia nodulifera ²	Mycetophyllia sp.	Telesto fruticulosa ²
Diodon holocanthus ¹	Mycteroperca acutirostris ¹	Tetraodontidae ¹
Diodon hystrix ¹	Mycteroperca bonaci ¹	Thalassoma bifasciatum ¹
Diodon sp. ¹	Mycteroperca interstitialis ¹	Thick turf
Diplastrella megastellata	Mycteroperca microlepis ¹	Thyroscyphus marginatus
Diplectrum bivittatum ¹	Mycteroperca phenax ¹	Thyroscyphus ramosus
Diplectrum formosum ¹	Mycteroperca tigris ¹	Tigrigobius macrodon ¹
Diplodus argenteus ¹	Mycteroperca venenosa ¹	Tigrigobius saucrus ¹
Diplodus holbrookii ¹	Myrichthys breviceps	Trachinotus falcatus ¹
Diploria labyrinthiformis ²	Myrichthys ocellatus	Trachinotus goodei ¹
Doratonotus megalepis ¹	Myripristis jacobus ¹	Trachurus lathami ¹
Echeneis naucrates	Narcine bancroftii	Trididemnum solidum
Echeneis neucratooides	Needlefish sp.	Turf algae free of sediment
Echidna catenata ¹	Negaprion brevirostris	Turf algae with sediment
Echinoidea	Neofibularia nolitangere	Tylosurus crocodilus
Ectyoplasia ferox	Neoniphon marianus ¹	Ulaema lefroyi
Elacatinus dilepis ¹	Nes longus ¹	Umbrina coroides ¹
Elacatinus evelynae ¹	Nicholsina usta ¹	Umimayanthus parasiticus
Elacatinus horsti ¹	Niphates digitalis	Undaria sp.
Elacatinus oceanops ¹	Niphates erecta	Unidentified fish
Elacatinus randalli ¹	Octocorallia ²	Unidentified species
Elacatinus xanthiprora ¹	Oculina diffusa ²	Unknown encrusting sponge
Elagatis bipinnulata ¹	Oculina sp. ²	Unknown rope sponge
Elops saurus	Ocyurus chrysurus ¹	Unknown substrate
Emblemaria pandionis ¹	Odontoscion dentex ¹	Unknown tube sponge
Emblemariopsis bahamensis ¹	Ogocephalus nasutus ¹	Upeneus parvus ¹
Emmelichthyops atlanticus ¹	Ogocephalus sp.	Urobatis jamaicensis
Enchelycore carychroa ¹	Oligoplites saurus ¹	Valonia macrophysa
Enchelycore nigricans ¹	Ophichthus ophis	Valonia ventricosa
Encrusting gorgonian ²	Ophioblennius macclurei ¹	Xyrichtys martinicensis ¹
Enneanectes altivelis	Opisthonema oglinum	Xyrichtys novacula ¹
Enneanectes boehlkei	Opistognathus aurifrons	Xyrichtys splendens ¹
Enneanectes jordani	Opistognathus macrognathus	Zebrasoma velifer ¹
Epinephelus adscensionis ¹	Opistognathus maxillosus ¹	Zoanthidae
Epinephelus drummondhayi ¹	Opistognathus sp.	Abudefduf saxatilis ¹
Epinephelus guttatus ¹	Opistognathus whitehursti	Abudefduf taurus ¹

1 - Coral Reef - Species Richness, 2 - Coral Reef - Percent Cover

References

1. Florida Department of Environmental Protection (DEP). [Florida STORET / WIN](#). (2024).
2. Florida Department of Environmental Protection (DEP); Office of Resilience and Coastal Protection (RCP); Coral Reef Conservation Program. [Southeast Florida Coral Reef Initiative \(SEFCRI\) Water Quality Monitoring Report](#). (2011).
3. Volusia County (Florida); Marine Discovery Center. [Lagoon Watch \(Formerly Marine Discovery Center\)](#). (2023).
4. Florida Fish and Wildlife Conservation Commission (FWC); Florida Fish and Wildlife Research Institute (FWRI). [Harmful Algal Bloom Marine Observation Network](#). (2018).
5. Florida Fish and Wildlife Conservation Commission (FWC). [Fisheries-Independent Monitoring \(FIM\) Program](#). (2022).
6. Florida Department of Environmental Protection (DEP); Office of Resilience and Coastal Protection (RCP); Coral Reef Conservation Program. [Southeast Florida Water Quality Assessment Survey](#) . (2024).
7. National Oceanic and Atmospheric Administration (NOAA); National Data Buoy Center. [National Data Buoy Center](#). (2024).
8. Florida Fish and Wildlife Conservation Commission (FWC); Florida Wildlife Research Institute (FWRI). [Water Temperature on Coral Reefs in the Florida Keys](#). (2022).